

Sujeet Singh

SENIOR SOFTWARE DEVELOPER · AI ENGINEER

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Summary

Backend engineer with 6+ years building production Python systems, including distributed pipelines, high-throughput APIs, and PostgreSQL at scale. Built patent intelligence platforms serving 100K+ monthly users over a 10M+ corpus, cutting data processing latency by 40%. Takes platforms from greenfield to launch, owning multi-tenant RBAC design and serverless ETL on AWS (Lambda, S3). Ships LLM-powered products into production: schema-aware natural-language search over patent and logistics data, and classification at 90%+ accuracy.

Work Experience

Hexalog

Gurugram, India

SDE-2

Mar 2026 - PRESENT

- Built the management reporting layer consolidating shipments, warehouse, sales, and resource-planning data from four disconnected ERP systems into a single queryable store, then layered a LangGraph agent on top that resolves ambiguous leadership questions into validated SQL across all source schemas.
- Architecting a global supply chain disruption monitoring system on a LangGraph event correlation pipeline that ingests news, port and vessel, weather, and geopolitical feeds, scoring disruption risk in real time and triggering automated alerts on affected shipments.

GreyB Analytics

Mohali, India

UTILITY LEAD - PATENT LITIGATION ANALYTICS

Oct 2025 - Mar 2026

- Drove the patent litigation analytics platform from an unaddressed market gap to 100K+ monthly users, architecting a FastAPI/PostgreSQL backend, Nuxt 3 frontend, and AWS infrastructure (S3, Lambda), then owning its scale and data pipeline reliability.
- Built an LLM-powered natural language search system with schema-aware SQL generation and constraint validation, enabling non-technical users to query complex patent datasets without hallucination risk in production.
- Integrated Claude and ChatGPT into patent classification pipelines via prompt engineering, automating categorization by type (Molecular, Formulation, Process) and drug portfolio mapping, achieving 90%+ accuracy across multilingual patent datasets.

SENIOR SOFTWARE DEVELOPER

Oct 2023 - Sep 2025

- Led development of a global drug and patent monitoring platform from design through launch, designing a multi-tenant architecture with RBAC to enable pharmaceutical clients to track all FDA-approved drugs and associated patent landscapes across global jurisdictions.
- Engineered asynchronous ingestion across 10+ global patent offices, using direct API integration where available and custom scrapers, normalizing every source into a single schema over a 10M+ patent corpus.
- Built a queue-based alert system delivering portfolio notifications with zero message loss on 10K+ actively monitored drug patents, cutting data processing latency by 40% by replacing scheduled polling with event-driven triggers.

SOFTWARE DEVELOPER

Jun 2020 - Sep 2023

- Built the FDA Orange Book ingestion pipeline covering drugs, active ingredients, patents, and exclusivities, computing deltas month over month to drive portfolio change notifications in the monitoring platform.
- Replaced manual analyst lookups with a serverless Python/Playwright pipeline on AWS Lambda, scraping several patent office sites into a single PostgreSQL schema of legal status, expiry, and family continuity, delivering 10x the throughput of the manual process with fewer data errors.
- Designed the PostgreSQL schema behind patent search, modeling patents, assignees, and legal status for query performance, then tuned indexes and plans to cut response times by 40%.

Technical Skills

Programming Languages

Python, JavaScript, SQL, C/C++

Backend & Infrastructure

FastAPI, Flask, Django, Celery, Node.js, REST APIs, AWS (Lambda, S3, EC2), Docker, CI/CD

Databases & Data

PostgreSQL, MySQL, Redis, indexing & query optimization, ETL pipelines, web scraping (Playwright)

AI & LLMs

LangGraph, Claude & OpenAI SDK, Ollama (local models), prompt engineering, natural-language-to-SQL, AI Agents

Frontend

Vue.js, Nuxt 3, HTML, CSS, Tailwind CSS

Education

Indian Institute of Information Technology, Una

Himachal Pradesh, India

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (CGPA: 8.17)

Jul 2016 - May 2020

Computer Science, Data Structures and Algorithms, OOP, DBMS, OS